Manit Sarkar

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Professional Summary

Mechanical Engineering undergraduate at Thapar Institute of Engineering and Technology with a **CGPA of 8.86/10**, awarded a **merit scholarship** for academic excellence and passionate about bridging the gap between **mechanical design, space systems & astrophysics engineering**. Hands-on experience applying ML to data pipelines and orbital dynamics, with strong proficiency in Python based ML frameworks (TensorFlow, PyTorch, Scikit-learn) and engineering simulation tools (CREO, MATLAB). Actively engaged in research on multiwavelength galaxy imaging and satellite trajectory prediction, with a long term goal of contributing to real world space science problems.

Education

Thapar Institute of Engineering and Technology (TIET) — Patiala, India

B.E. Mechanical Engineering (6th Semester, 3rd Year) - CGPA: **8.86 / 10** — *Expected: June 2027*

- Awarded **Merit Scholarship (50% tuition fee waiver)** for academic performance in the 2nd year of B.Tech.

Internship

Experiential Learning (EL) Summer Internship | GitHub — June – July 2025

Thapar Institute of Engineering and Technology | Mentor: Dr. Mamta Gulati

- Classified galaxy morphologies (spirals, ellipticals, irregulars) from a 31,374-sample dataset using both galaxy images and photometric measurements (brightness, magnitude, redshift, stellar mass, star formation rate).
- Combined predictions from a **ResNet-18 CNN** (image-based) and **XGBoost** (numerical features) by training a blending model on their outputs, achieving **∼93% overall accuracy** (elliptical: 94%, spiral: 92%, irregular: 92%).

Research & Projects

Multiwavelength Galaxy Morphology Classification using AI/ML — *Jan 2025 – Present*

Student Research | Thapar Institute of Engineering and Technology | Mentor: Dr. Mamta Gulati

- Expanding on EL internship work by developing a **CNN-based pipeline** to classify galaxies using multiwavelength (more than 3 wavelengths) imaging data (as opposed to the numerical/tabular approach used previously).

Orbital Dynamics & Satellite Trajectory Prediction using ML | GitHub — *Sep 2025 – Present*

Student Research | Thapar Institute of Engineering and Technology | Mentor: Dr. Mamta Gulati

- Investigating how orbital perturbations (solar radiation pressure, atmospheric drag, gravitational harmonics) affect satellite trajectories and using **ML to predict trajectory deviations during solar storm events**.

Flow-Induced Vibration Wind Energy Harvester — *Jan 2026 – Dec 2026*

B.E. Capstone Project | Thapar Institute of Engineering and Technology | Mentor: Dr. Ashish Purohit

- Designing a **flow-induced vibration energy harvester**, testing multiple bluff body geometries across bending, torsional, and **VIV–galloping hybrid** flow regimes via wind tunnel experiments.

Certifications

View All Certificates

Machine Learning Specialization (3-courses) — June 2024

DeepLearning.AI & Stanford University | Andrew Ng | Coursera

- Supervised learning | Unsupervised learning | Recommender systems and Reinforcement learning | Advanced learning algorithms

TensorFlow Developer Certificate Courses (4 courses) — Jul – Aug 2024

DeepLearning.AI | Laurence Moroney / Andrew Ng | Coursera

- Intro to TensorFlow for AI, ML & Deep Learning (Jul 2024) | CNNs in TensorFlow (Jul 2024) | NLP in TensorFlow (Aug 2024)

Deep Learning Specialization (1 course) — Sep 2024

DeepLearning.AI & Stanford University | Andrew Ng | Coursera

- Neural Networks & Deep Learning

Technical Skills

Programming	Python (primary), C / C++
ML / AI	Supervised & Unsupervised Learning, Computer Vision, Neural Networks, NLP, Image Processing
Libraries	TensorFlow, PyTorch, Scikit-learn, NumPy, Matplotlib, OpenCV, MediaPipe, YOLO, Astropy, AstroQuery, SpiceyPy
CAD / Sim	SolidWorks, CREO Parametric, OnShape, AutoCAD, MATLAB
Languages	English (Fluent), Hindi (Native), French (Beginner)